

End-only + Mylar-wrapped EJ-204 bar:
Optical simulation study
Geant4 11.4.0 / SSLG4-OPSim

R. Ríos (ULS)

SHiP Collaboration Meeting

2026-06-14

Motivation & scope

- ▶ **Deliberate simplification:** readout *only* at the two ends ($\pm X$ faces); all other faces wrapped in Mylar.
- ▶ **Goals:**
 - ▶ Isolate end-readout contribution.
 - ▶ Characterize multi-component attenuation with Mylar.
 - ▶ Benchmark the SUM4 timing estimator on a clean geometry.
- ▶ 31 beam positions ($x = -690 \dots +690$ mm), 2,000 μ^- events per position.

This study

End-only + Mylar = extreme limit. Photons either exit to an end SiPM or are absorbed by Mylar after multiple reflections.

No TB counterpart

This configuration has **no direct test-beam counterpart** (April-2026 TB used top readout + Tyvek).

Electronic jitter not included

SPTR $\oplus$ FastIC jitter are post-processing steps, not folded into σ_t shown here.

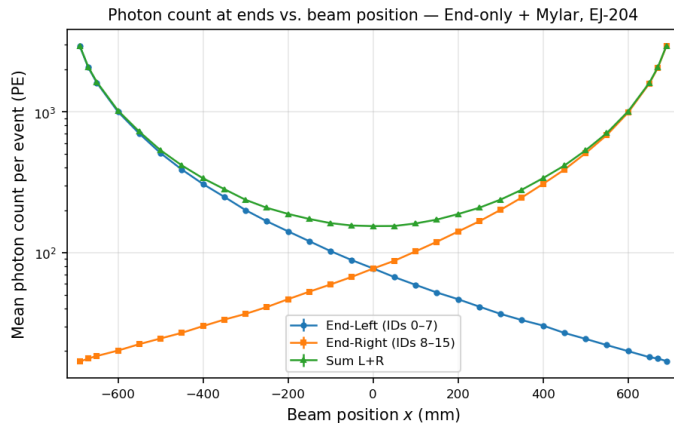
Geometry & simulation setup

- ▶ **Bar:** EJ-204 (OPSC-101), $1400 \times 60 \times 10 \text{ mm}^3$.
- ▶ **Longitudinal axis:** X (-700 to $+700 \text{ mm}$).
- ▶ **End-left SiPMs** (IDs 0–7): $-X$ face.
End-right SiPMs (IDs 8–15): $+X$ face.
- ▶ **Wrapping:** Mylar on $\pm Y$ and $\pm Z$ faces;
 $R = 0.90$, specular lobe = 1.0, $\sigma_\alpha = 0.1^\circ$.
- ▶ **Readout:** SUM4 leading-edge at 4 PE; groups
 $\{0-3\}/\{4-7\}$ (left), $\{8-11\}/\{12-15\}$ (right).
- ▶ **Estimator:** $\sigma_{\text{single}} = \sigma_{LR}/\sqrt{2}$.
- ▶ **Beam:** μ^- , 1 GeV, normal incidence.

Key parameters

Scint. yield	10,400 γ/MeV
Emission peak	408 nm
Bulk λ	160 cm
n	1.58
τ_r/τ_d	0.7 ns/1.8 ns
Mylar R	0.90

Photon yield at ends vs. beam position



Characteristic values

Near left end ($x = -690$ mm):

Left ≈ 2926 PE

Right ≈ 2914 PE

Center ($x = 0$):

$\langle N_{pe}^L \rangle = 77.2$ PE

$\langle N_{pe}^R \rangle = 77.0$ PE

Observation

Log-scale reveals two regimes: prompt near-end region (steep) and recirculated floor (~ 77.2 PE at center). Mylar recirculates but also absorbs.

Two attenuation lengths: datasheet vs. system measurement

$\lambda_{\text{bulk}} = 160 \text{ cm}$ (datasheet)

Material property. Loaded as ABSLENGTH in `Materials.cc:99`. Probability of straight-line absorption in the plastic. **Knows nothing** about bar geometry or reflectors.

λ_{eff} (what we fit)

System observable: slope of NPE vs. distance at end. Combines bulk absorption *and* geometry *and* reflection losses. Always $\lambda_{\text{eff}} < \lambda_{\text{bulk}}$. **These two quantities are not directly comparable.**

Why a single exponential fails

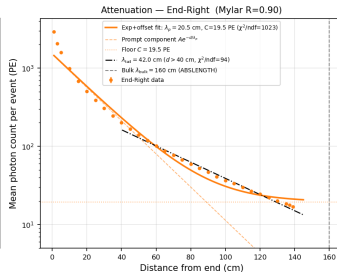
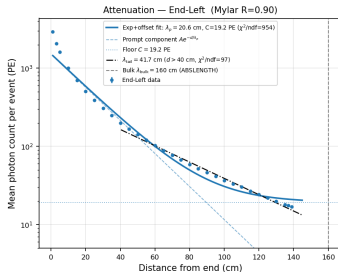
Three numbers that should agree if $N_0 e^{-d/\lambda}$ held:

1. Two-point ($d=10 \text{ mm} \rightarrow 700 \text{ mm}$):
 $\lambda \approx 19 \text{ cm}$
2. Single-exp fit: $\lambda \approx 32.2 \pm 1.9 \text{ cm}$,
 $N_0 = 811 \text{ PE}$
3. Actual NPE at $d \approx 0$: 2926 PE

Items (2) and (3) disagree by $\approx 3.6\times$.
 $\chi^2/\text{ndf} = 2321$ confirms the model is wrong.

Physical reason: the NPE profile has two components — prompt (short-range) and recirculated floor (Mylar-sustained).

Attenuation: two-component fit and tail λ



Exp+offset model (left side)

$$N(d) = A e^{-d/\lambda_p} + C$$

$$\lambda_p = 20.6 \pm 1.1 \text{ cm (prompt)}$$

$$C = 19.2 \pm 2.0 \text{ PE (recirculated floor)}$$

$$\chi^2/\text{ndf} = 954$$

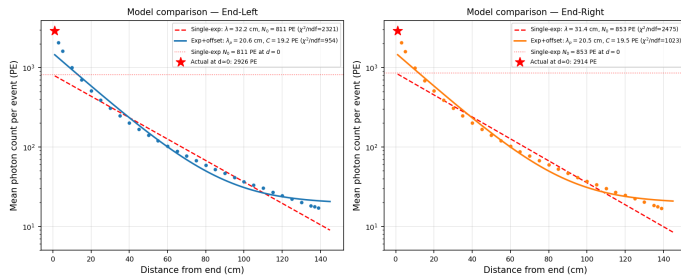
Tail fit ($d > 40$ cm):

$$\lambda_{\text{tail}} = 41.7 \pm 0.9 \text{ cm}$$

$$\lambda_{\text{eff}} \neq \lambda_{\text{bulk}}$$

160 cm (bulk) is the material ABSLENGTH. $\lambda_p \approx 21$ cm and $\lambda_{\text{tail}} \approx 42$ cm are *system* observables: dominated by geometry + Mylar losses, not bulk absorption.

Single-exp vs. exp+offset: model comparison



Key diagnostics

Model	λ (cm)	χ^2/ndf
Single-exp	32.2 ± 1.9 cm	2321
Exp+offset	20.6 ± 1.1 cm	954
$\lambda_{\text{tail}} (d > 40 \text{ cm})$	41.7 ± 0.9 cm	96.5

Inconsistency proof

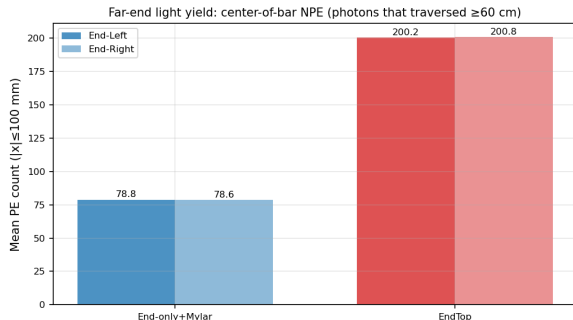
Red star: actual NPE at $d \approx 0$ is 2926 PE.

Single-exp $N_0 = 811$ PE (red dashed).

Factor ≈ 3.6 inconsistency.

Single-exp $\chi^2/\text{ndf} = 2321$ confirms failure.

Far-end light yield: where reflector quality matters



Center-of-bar NPE ($|x| \leq 100$ mm)

Config	R	NPE
End-only+Mylar	0.90	77.2 PE
EndTop	0.98	194.4 PE

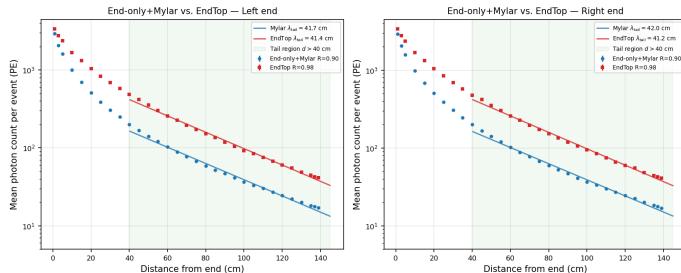
Ratio EndTop/Mylar $\approx 2.5\times$.

Physical interpretation

Both configs see photons after ≥ 60 cm propagation and many reflections. At $R = 0.90$ (Mylar), each bounce absorbs 10%: cumulative loss dominates. At $R = 0.98$, only 2% per bounce. **Reflector**

quality, not λ_{bulk} , controls the recirculated light level.

Apples-to-apples: End-only+Mylar vs. EndTop



Identical pipeline

Both datasets processed with the **same** end-SiPM-only selection, attenuation models, and tail-fit range.

λ_{tail} comparison ($d > 40$ cm)

Config	λ_{tail}
End-only+Mylar	41.7 ± 0.9 cm
EndTop	41.4 ± 0.7 cm

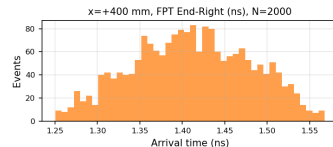
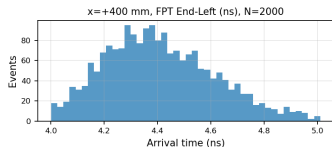
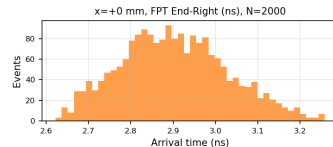
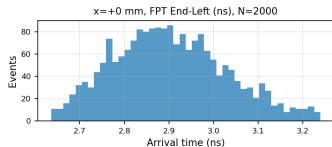
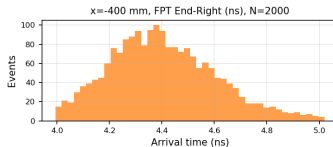
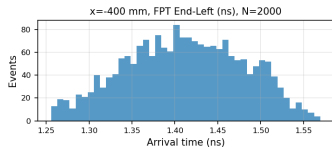
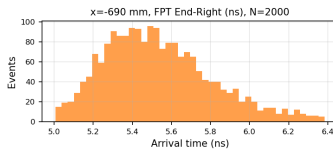
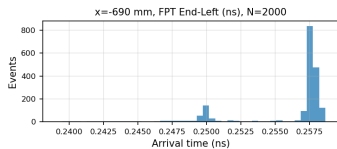
Similar slope — dominated by geometry, not reflector type.

Conclusion

λ_{tail} does *not* distinguish configs; the NPE *amplitude* does.

End-only+Mylar has $\approx 2.5 \times$ less light at center due to $R = 0.90$ vs $R = 0.98$.

Arrival-time distributions: FPT at representative positions



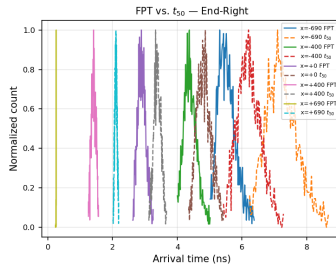
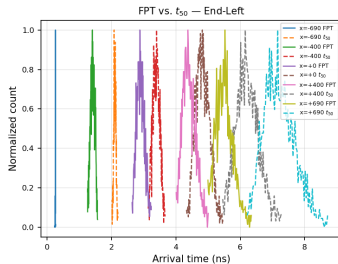
Distributions shown

First Photon Time (FPT) at End-Left and End-Right per event (**not normalized**). Positions: $x \in \{-690, -400, 0, +400, +690\}$ mm. $N = 2,000$ events per position.

Key feature

Distribution shifts to larger t and broadens as beam moves away from the respective end — longer propagation path and fewer photons (statistical minimum effect).

FPT vs. t_{50} : comparing timing estimators



FPT vs. t_{50}

FPT: minimum of photon arrival times per event.

t_{50} : median — less sensitive to the N -dependent statistical minimum. t_{50} systematically later

than FPT; offset grows with distance.

v_{eff} bias

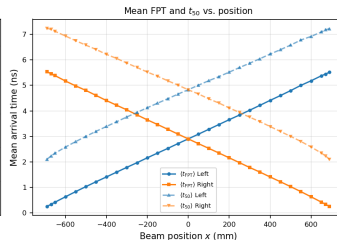
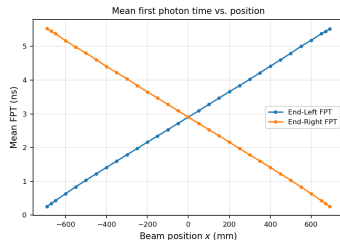
FPT $v_{\text{eff}} = 264.1 \text{ mm/ns}$

(apparent, biased by N -dependent statistical minimum).

SUM4 Δt gives

$v_{\text{eff}} = 213.7 \text{ mm/ns}$ (robust).

Mean arrival time and effective velocity



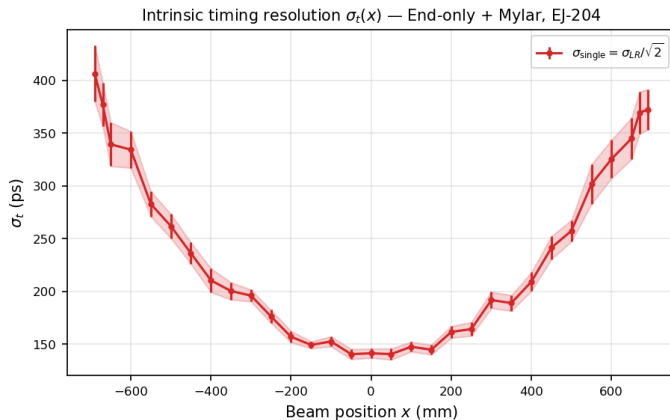
v_{eff} summary

Estimator	v_{eff}
SUM4 Δt	213.7 mm/ns
FPT slope	264.1 mm/ns
t_{50} slope	275.7 mm/ns

FPT anomaly

$v_{\text{eff}}^{\text{FPT}} > c/n$ ($c/n \approx 190$ mm/ns for $n = 1.58$). Artifact: FPT = statistical minimum of N photon arrival times; fewer photons at large $d \rightarrow$ earlier FPT bias.

Intrinsic timing resolution $\sigma_t(x)$



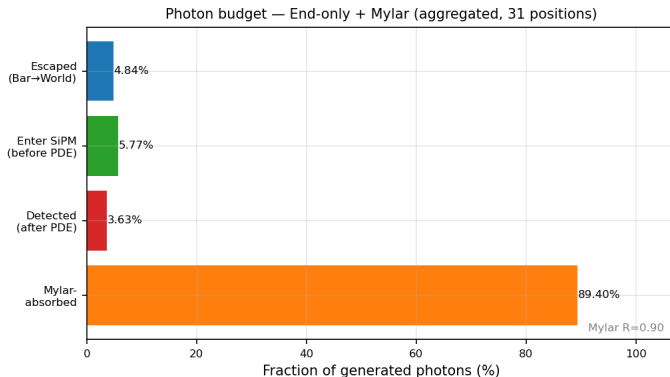
Key values

Position	σ_{single}
$x = -690$ mm	406 ± 27 ps
$x = -400$ mm	211 ± 11 ps
$x = 0$	142 ± 4 ps
$x = +400$ mm	209 ± 9 ps
$x = +690$ mm	372 ± 19 ps

Intrinsic only

σ_t = optical + SUM4 estimator.
SPTR (≈ 106 ps) and FastIC
(≈ 10 ps) not folded in.

Photon budget — End-only + Mylar



Fractions

Fate	Fraction
Mylar-absorbed	$\approx 4.84\%$
Enters SiPM	5.77%
Detected (PDE)	3.63%

Mylar loss dominates

Each Mylar bounce:

$(1 - R) = 0.90$ absorbed. $\Rightarrow$

Over many bounces in a thin bar, cumulative loss $\gg$ bulk loss. This explains $\lambda_p \approx 21 \text{ cm} \ll 160 \text{ cm}$.

Summary

Key results

- ▶ **Two attenuation lengths:** $\lambda_{\text{bulk}} = 160 \text{ cm}$ (ABSLENGTH, material property); $\lambda_p \approx 21 \text{ cm}$ (prompt system component); $\lambda_{\text{tail}} \approx 41.7 \pm 0.9 \text{ cm}$ (tail, comparable metric). These are *not* the same quantity.
- ▶ **Single-exp fails:** $\chi^2/\text{ndf} = 2321$; N_0 inconsistency factor $\approx 3.6\times$. Two-component model required.
- ▶ **Reflector R controls tail NPE:**
End-only+Mylar ($R = 0.90$): 77.2 PE at center;
EndTop ($R = 0.98$): 194.4 PE. Ratio $\approx 2.5\times$.
- ▶ λ_{tail} is *equal* in both configs — the NPE amplitude is the distinguishing observable.
- ▶ $\sigma_t^{\text{intr.}}$: $142 \pm 4 \text{ ps}$ (center) to $406 \pm 27 \text{ ps}$ (ends) — intrinsic only.

Next steps

1. Fold in electronic jitter:
 $\sigma_{\text{SPTR}} = 106 \text{ ps} \oplus \sigma_{\text{FastIC}} = 10 \text{ ps}$.
2. Leading-edge time-walk correction.
3. Scan Mylar $R \in [0.85, 0.95]$ to map NPE yield vs. reflector quality.
4. Scan with Tyvek for direct comparison to April-2026 TB configuration.

Appendix A: simulation parameters

Parameter	Value	Unit	Source
Bar material	EJ-204 (OPSC-101)		SSLG4/Materials.cc
Bar dimensions	$1400 \times 60 \times 10$	mm^3	GDML
Readout	End-only ($\pm X$ faces)		geometry
Wrapping ($\pm Y, \pm Z$)	Mylar		run_metadata
Mylar reflectivity R	0.90		run_metadata
Mylar specular lobe	1.0		run_metadata
Mylar σ_α	0.1	deg	run_metadata
Scint. yield	10,400	γ/MeV	Eljen EJ-204 datasheet Aug 2023; Materials.cc:100
Emission peak	408	nm	Eljen EJ-204 datasheet Aug 2023; Materials.cc:86
Bulk λ (ABSLENGTH)	160	cm	Eljen EJ-204 datasheet Aug 2023; Materials.cc:99
Refractive index n	1.58		Eljen EJ-204 datasheet; Materials.cc:41
Density	1.023	g/cm^3	Eljen EJ-204 datasheet; Materials.cc:24
Rise time τ_r	0.7	ns	SSLG4 (EJ-204)
Decay time τ_d	1.8	ns	SSLG4 (EJ-204)
SiPM model (physical)	AFBR-S4N66P014M		hardware
SiPM PDE (peak)	63	% @ 420nm	datasheet
Beam	μ^- , 1 GeV		macro
Positions	31		run_metadata
Events/position	2,000		run_metadata
Geant4 version	11.4.0		log

Appendix B: σ_t per-position summary

x (mm)	N_{trig}	Trig. eff.	σ_{single} (ps)	Err. (ps)
−690	1911	0.956	406.2	26.6
−400	2000	1.000	210.6	11.2
+0	2000	1.000	141.5	4.3
+400	2000	1.000	209.3	8.9
+690	1911	0.956	372.1	19.1

All values in ps. $\sigma_{\text{single}} = \sigma_{LR}/\sqrt{2}$ from SUM4 leading-edge core Gaussian fit.

Appendix C: attenuation fit details (three models)

Model	Side	N_0 or A (PE)	λ (cm)	C (PE)	χ^2/ndf
Single-exp	Left	811 ± 106	32.2 ± 1.9	—	2321
Exp+offset	Left	1502 ± 146	20.6 ± 1.1	19.2 ± 2.0	954
$\lambda_{\text{tail}} (d > 40 \text{ cm})$	Left	429 ± 21	41.7 ± 0.9	—	96.5
Single-exp	Right	853 ± 110	31.4 ± 1.8	—	2475
Exp+offset	Right	1501 ± 142	20.5 ± 1.1	19.5 ± 2.1	1023
$\lambda_{\text{tail}} (d > 40 \text{ cm})$	Right	423 ± 21	42.0 ± 0.9	—	93.6

All fits weighted by per-point SEM. Errors from covariance matrix, scaled by $\sqrt{\chi^2/\text{ndf}}$ when > 1 . Tail fit: $d > 40 \text{ cm}$ only.

Appendix D: method & caveats

- ▶ **Geometry:** Bar $1400 \times 60 \times 10 \text{ mm}^3$; long axis X . End SiPMs on $\pm X$; Mylar on $\pm Y, \pm Z$.
- ▶ **SUM4 trigger:** leading-edge at 4 PE. Groups $\{0-3\}/\{4-7\}$ (left); $\{8-11\}/\{12-15\}$ (right).
- ▶ σ_t **is intrinsic only:** optical + estimator; SPTR ($\approx 106 \text{ ps}$) and FastIC ($\approx 10 \text{ ps}$) not folded in.
- ▶ **SiPM naming:** AFBR-S4N66P024M is used *only* for the PDE curve; physical device is the single-element **AFBR-S4N66P014M**.
- ▶ **No TB counterpart:** April-2026 TB used top readout + Tyvek wrapping $\neq$ end-only + Mylar.
- ▶ **Longitudinal axis** = X (not Z or Y).
- ▶ **Yield:** OPSC-101 loaded with $10,400 \gamma/\text{MeV}$ in `Materials.cc:100` — matches Eljen EJ-204 datasheet Aug 2023.
- ▶ **EndTop reference:** EXEC_07, 2,000 events/position, same 31 positions. Side reflectors $R = 0.98$ (BarSkinReflector), NOT Mylar.